

MUHAMMAD SAMMAR ISLAM

Electrical and Electronic Engineer

☎ +966 580681209 ✉ samarislam678@gmail.com  [Linkedin](#)

PERSONAL INFORMATION

- Date of Birth: 08-Sep-2000
- Gender: Male
- Marital Status: Single
- Nationality: Pakistani
- Address: 18 AL Kharj Road, Riyadh, Saudi Arabia
- Iqama: Transferable iqama

OBJECTIVE

Motivated Electrical Engineering undergraduate from GIKI with hands-on experience in industrial automation, circuit design, IoT systems, PCB prototyping. Skilled in Python, MATLAB, and embedded platforms including Arduino and PIC microcontrollers. Eager to contribute to innovative engineering teams through strong problem-solving, adaptability, and a passion for integrating hardware with intelligent systems.

EXPERIENCE

Intern at Pakistan Oil Field Limited (POL)

June 2023 – July 2023

Role- Project Engineer

- Learned about power generation, transmission, and distribution, including POL's independent 3.2 MW electricity generation system.
- Gained knowledge of motors, power transformers, and industrial distribution systems.
- Explored instrumentation systems, focusing on industrial sensors and their applications.

Intern at Pakistan Aeronautical Complex Kamra (PAC)

July 2024 – July 2024

Role- Design Engineer

- Gained experience at Pakistan Aeronautical Complex, a leading defense contractor in aerospace and military support.
- Contributed to the design, development, and testing of aircraft and avionics systems.
- Assisted in troubleshooting and maintaining avionics systems.

Intern at Creative Electronics and Automation (Crest)

July 2024 – August 2024

Role- Product Design Engineer

- Creative Electronics and Automation (Crest) is Automation Solutions, Manufacturing, and Export of Electronics Products and Industrial IT Services.
- Crest being the only LED Lights manufacturer in Pakistan having complete automatic LED assembly line with international level testing facility.

PROJECTS

FYP Rotary Inverted Pendulum like a Quanser Qube-Servo 3 | PID Control, MATLAB, Simulink 2025

- Designed and implemented a rotary inverted pendulum system inspired by the Quanser Qube Servo 3.

- Developed a mathematical model, including dynamics and state-space representation.
- Implemented PID control for stabilization and swing-up using MATLAB/Simulink and embedded hardware.

Active Noise Cancellation System (ANC) | Communication, Signal, Python *2024*

- Active noise cancellation can be applied to various situations, such as headphones and earplugs.
- It is important to design and evaluate an active noise cancellation system that can achieve a high noise reduction and quality of sound.

Starter For Three Phase Induction Motor | STAR-DELTA, TPDT *2024*

- A device that provides the initial torque to start the motor.
- It is used to control the starting current of the motor and protect it from overloading.

Solar Inverter Without Battery | AC,DC *2023*

- Solar inverters without batteries convert the DC generated by solar panels into AC for use in homes.
- In this setup, the energy produced is either consumed directly or sent back to the grid, eliminating the need for energy storage.

Home automation Using Pic Microcontroller | Altium Designer, PCB *2023*

- This is a system that can remotely control AC home appliances using an IR remote.
- The project can be designed on a PCB using Altium designer.

Water Level Indicator | Arduino, Sensor *2022*

- This is a circuit that can be built using an Arduino board and a water sensor module.
- The circuit is designed to indicate three levels of water stored in the tank.

EDUCATION

Ghulam Ishaq Khan Institute Science and Technology (GIKI) *2021 – 2025*

Bs - Electrical and Electronic Engineering - **CGPA - 2.88** *Topi, Sawabi*

FSC (Pre-Engineering) *2019 – 2020*

Cambridge International College Talagang (CISC), Chakwal

Matriculation (Science) *2017 – 2018*

Cambridge International College Talagang (CISC), Chakwal

SKILLS

Languages:

- Python
- C++
- Assembly for PIC Microcontroller
- Data Structures and Algorithms (DSA)

Technologies/Frameworks:

- Altium Designer

- PLC
- AutoCAD
- Digital Signal processing (DSP)
- Digital Logic Design (DLD)

Developer Tools:

- MATLAB/Simulink
- Protues Software
- CREO
- MPLAB
- LTSpice and PSpice
- LabVIEW
- Multisim

Digital Skills:

- Microsoft Office
- Microsoft Word
- Microsoft Excel
- Outlook
- power point

EXTRACURRICULAR

- Member of “Institution of Engineering and Technology” (IET).
- Participated in several School competitions (DEBATE (English, Urdu)).
- Head for SUMMIT event of IET.
- Organized ‘STUDENT GALA’ (2020) and ‘SCIENCE FAIR’.
- Member of “GIKI SPORTS SOCIETY” (GSS).
- Winner of the Intercampus sports event (2019-2020)

LANGUAGES

English, Urdu, Punjabi